

again clipped at 0 for very large discrepancies. When a value is non-numeric, s_{val} falls back to a textual similarity term using NL_{τ} . The overall similarity between two mappings is then

$$s(i, j) = s_{\text{key}}(i, j) s_{\text{val}}(i, j), \quad (5)$$

which is close to 1 only if both the key (row+column) and the associated value match well.

To compare a predicted table $P = \{p_i\}_{i=1}^N$ with a gold table $T = \{t_j\}_{j=1}^M$, RMS first constructs an $N \times M$ cost matrix based on the key similarity and solves a minimum-cost bipartite matching problem to obtain a binary assignment matrix $X \in \{0, 1\}^{N \times M}$. Using this matching, RMS precision and recall are defined as

$$\text{RMS}_{\text{precision}} = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M X_{ij} s(i, j), \quad (6a)$$

$$\text{RMS}_{\text{recall}} = \frac{1}{M} \sum_{i=1}^N \sum_{j=1}^M X_{ij} s(i, j), \quad (6b)$$

$$\text{RMS}_{\text{F1}} = \frac{2 \cdot \text{RMS}_{\text{precision}} \cdot \text{RMS}_{\text{recall}}}{\text{RMS}_{\text{precision}} + \text{RMS}_{\text{recall}}}. \quad (6c)$$

Because RMS treats tables as unordered sets of (row, column, value) mappings and considers both the original and transposed layouts, it is by construction invariant to row and column permutations and to table transpositions.

In our experiments, for each query we take the SPARQL result table as the gold table T and the output of each neural system (neural-only and symbiotic context-augmented neural settings) as the predicted table P . We compute RMS precision, recall, and F1 using the original DePlot implementation’s default thresholds (for textual and numeric similarity), as exposed via our evaluation script. We report per-query RMS scores as well as macro-averaged RMS across queries for each system and evaluation setting, thereby quantifying both how many of the gold rows and cells are recovered (recall) and how many of the predicted entries genuinely correspond to SPARQL-derived evidence (precision).

C Results – Detailed analysis of domain expert impressions (RQ1)

Here we provide a more detailed account of the expert evaluation results for RQ1. This was based on the survey taken by 3 domain experts with the following outcome.

Meaningfulness. Across all 33 queries, the questions themselves were judged to be scientifically well-posed. The mean meaningfulness score was 4.54/5, with 96% of all ratings at ≥ 4 , and 32 out of 33 queries achieving an average meaningfulness score of at least 4. ALE queries were rated slightly higher (mean 4.69) than ALD queries (mean 4.42), and all ALE ratings fell in the range 4–5. As visualised in Fig. 7 (see also the per-domain panels in Figs. 7a and 7b), individual per-query, per-annotator points cluster tightly in the upper part of the scale, with only a few isolated ratings below 3 (e.g., for ALD Q11). This indicates that the query set, as a whole, aligns well with the experts’ sense of plausible relationships, appropriate concepts, and relevant use cases in ALD and ALE.

Among the ALD queries, six achieved perfect mean scores of 5.00 (Q09, Q10, Q13, Q14, Q17, Q18), while the lowest-rated query was Q11 with a mean of 3.67—the only query across either domain to fall below 4.00. For ALE, four queries achieved perfect scores of 5.00 (Q02, Q05, Q07, Q08), and the lowest-rated queries (Q10, Q12, Q13) still attained a mean of 4.33. At the individual annotator level, mean ratings ranged from 4.21 to 5.00 for ALD and from 4.29 to 4.93 for ALE, reflecting moderate variation in stringency across evaluators.

Formal inter-annotator agreement, measured via Fleiss’ κ and Krippendorff’s α_{ordinal} , was low (ALD: $\kappa = 0.021$, $\alpha_{\text{ordinal}} = 0.081$; ALE: $\kappa = -0.114$, $\alpha_{\text{ordinal}} = -0.088$). This is a known artefact when ratings are strongly skewed toward one end of the scale: when nearly all scores are 4 or 5, even small disagreements (e.g., 4 vs. 5) can produce low or negative κ values despite broad substantive consensus. In our case, the high proportion of ratings at 4–5, combined with high exact-match rates (35.7%–78.6% across annotator pairs) and full three-way agreement on 28.6% (ALE) to 36.8% (ALD) of queries, suggests that annotators agreed that the queries were meaningful, even if they differed slightly in how enthusiastically they expressed this via the 5-point scale.

Usefulness. Experts also rated the usefulness of the SPARQL result tables as synthesized knowledge artefacts. Over all queries, the average usefulness score was 4.08/5, with 69% of ratings at ≥ 4 and more than half at the maximum value of 5. On a per-query basis, 25 of 33 queries achieved a mean usefulness score of at least 4, and only two queries received an average usefulness below 3. ALD queries received a marginally higher usefulness score (mean 4.16) than ALE queries (mean 3.98). Figure 8 (panels 8a and 8b) shows that usefulness ratings are somewhat more dispersed than meaningfulness ratings, especially for a small subset of queries whose tables are sparse or missing key parameters, but the majority of points still lie in the 4–5 band.

Among ALD queries, twelve achieved perfect mean usefulness scores of 5.00 (Q01, Q02, Q04, Q05, Q06, Q13, Q14, Q15, Q16, Q17, Q18, Q19), while the lowest-rated query was Q08 with a mean of 3.00. For ALE, the highest-rated query was Q06 with a mean of 4.67, and the lowest-rated queries were Q12 (mean 2.67) and Q14 (mean 3.00)—the only two queries across both domains to fall below 3.00. At the individual annotator level, mean usefulness ratings ranged from 4.21 to 5.00 for ALD and from 2.93 to 4.36 for ALE, indicating greater variation in evaluator stringency for the usefulness criterion than for meaningfulness.

Formal inter-annotator agreement was again low: for ALD, Fleiss’ $\kappa = -0.035$ and $\alpha_{\text{ordinal}} = 0.018$; for ALE, $\kappa = 0.151$ and $\alpha_{\text{ordinal}} = -0.041$. In the ALD case, these values are largely driven by a ceiling effect: two of three annotators assigned maximum scores to nearly all queries, so there is little variability for the statistics to capture, even though 63.2% of queries show full three-way agreement. ALE usefulness ratings exhibit greater dispersion, with one annotator providing systematically lower scores (mean 2.93) than the other two (means 4.36 and 4.21). As a result, full three-way agreement drops to 21.4%. Pairwise agreement between the two higher-scoring annotators remains substantial (78.6% exact match, Cohen’s $\kappa = 0.576$), suggesting that low overall agreement is driven mainly by different baseline expectations for what constitutes a “useful” synthesis, rather than inconsistent application of the criterion.

Taken together, these quantitative results suggest that domain experts not only regard the questions as meaningful but also find the machine-actionable syntheses produced by the ORKG comparisons genuinely helpful for reasoning about processes, mechanisms, and material–performance trade-offs.

Qualitative feedback. Free-text explanations for lower usefulness ratings point primarily to limitations in the current data modelling and coverage rather than to the concept of machine-actionable reviews itself. Experts frequently asked for additional parameters (e.g., sticking coefficients, surface orientation and termination, clearer precursor/co-reactant distinctions, explicit dopant columns, error bars or standard deviations on growth-per-cycle, and more detailed timing parameters per ALD/ALE cycle), more consistent categorisation of mechanisms and precursor families, and improved result presentation (e.g., removing duplicate rows, compacting very long tables, or explaining derived indices such as the efficiency metric in the rare-earth MOSLED example). In several cases, low scores were attributed to incomplete or sparsely populated ORKG comparisons, or to queries that, while technically correct, remained too generic to support concrete decision-making. This pattern indicates that dissatisfaction is concentrated on specific modelling choices and missing annotations, not on the idea of querying symbolic review knowledge.

The open-ended questions corroborate this interpretation and highlight both enthusiasm and pragmatic concerns. When asked to rate the overall usefulness of machine-actionable scientific knowledge, all three experts gave scores between 4 and 5 (mean 4.67), emphasising the value of faster access to synthesized insights and the ability to compare conditions across papers instantly instead of manually reconstructing spreadsheets from multiple reviews. Their “wish-list” questions extend this theme: examples include queries over the most common surface orientations and terminations for a given material, multi-parameter aggregations over ALD recipes grouped by precursor/co-reactant, temperature, plasma power, and reactor type, or systematic overviews of process windows and performance metrics for specific device architectures. In other words, experts immediately project the ORKG comparisons onto richer, multi-dimensional queries that mirror the way they already reason about processes and materials.

At the same time, experts foresee several barriers to broader adoption. These include (i) the additional time and effort needed to enter machine-actionable tables alongside traditional PDFs, (ii) uneven reporting practices in the literature, which limit what can be modelled consistently, (iii) concerns about data quality and trust if tables were to be (partially) auto-extracted, and (iv) the lack of established

community standards in ALD/ALE for which process parameters and performance metrics should be captured. On the positive side, they identify clear incentives that would motivate them to use or publish such tables: demonstrable time savings in literature review, community-wide standardisation of schemas, user-friendly tooling for authoring and querying ORKG comparisons, and clear benefits for their own workflows (e.g., being able to answer complex comparative questions “in one click”).